

Topology

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Chapter 1

Preliminaries

1.1 Axiom of Choice

Axiom of choice. Given a collection $\mathcal{A}$ of disjoint nonempty sets, there exists a set C consisting exactly one element from each element of $\mathcal{A}$.

Note. This means that the set C is such that C is contained in the union of the elements of $\mathcal{A}$ and for each $A \in \mathcal{A}$ the set $C \cap A$ has a single element.

Lemma 1.1 (Choice function). Given a collection $\mathcal{B}$ of nonempty sets (not necessarily disjoint), there exists a function,

$$c : \mathcal{B} \rightarrow \bigcup_{B \in \mathcal{B}} B$$

such that $c(B)$ is an element of B , for each $B \in \mathcal{B}$

Note. the function c is called a *choice function* for the collection $\mathcal{B}$

Remark. The difference from the A.O.C is that in this lemma the sets need not be disjoint. So we have $\mathcal{B}$ as the powerset of any set (excluding ϕ that is)

Proof. For a given B in $\mathcal{B}$ consider,

$$B' = \{(B, x) : x \in B\}$$

Now note that B' is nonempty as B is nonempty. Also note that for distinct B_1, B_2 we have B'_1, B'_2 are distinct as well. Now consider,

$$\mathcal{C} = \{B' : B \in \mathcal{B}\}$$

Note that sets in $\mathcal{B}$ are disjoint. So we can use axiom of choice there is a set c consisting of exactly one element from each elements of $\mathcal{C}$. Now note we have an assignment for each element of $\mathcal{C}$ to an ordered pair (B, x) and hence c is a rule for a function from the collection $\mathcal{B}$ to the set $\bigcup_{B \in \mathcal{B}} B$ $\square$

1.2 Well Ordered sets

Definition 1.2. A set A with an order relation $<$ is said to be well-ordered if every nonempty subset of A has a smallest element.

Example. $\{1, 2\} \times \mathbb{Z}_+$ in the dictionary ordering, so we have,

$$(1, 1), (1, 2), (1, 3), \dots, (2, 1), (2, 2), (2, 3)$$

where,

$$(a, b) < (c, d) \text{ if } a < c \text{ or } a = c \text{ and } b < d$$

◇

Example. The set of integers is not well-ordered in the usual sense. For instance the subset containing the negative integers has no smallest element. Neither is the real numbers in $[0, 1]$ either as any open subset does not have a smallest element. ◇

Theorem 1.3. Every nonempty finite ordered set has the order type of a section $\{1, \dots, n\}$ of $\mathbb{Z}_+$, so it is well ordered.

Proof. First we claim that every finite ordered set has a largest element (by induction). Secondly we can construct an order preserving bisection of A with $\{1, \dots, n\}$ for some n by induction as well. Trivial for $n = 1$, assume for $n - 1$, now for case n we isolate the largest element, use $n - 1$ case and define $f : A \rightarrow \{1, \dots, n\}$ by setting $f(x) = f'(x)$ and $f(b) = n$ where $f' : A - \{b\} \rightarrow \{1, \dots, n - 1\}$. □

Theorem 1.4. If A is a set, there exists an order relation on A that is a well-ordering

Chapter 2

Topological Spaces and Continuous Functions

2.1 Topological Spaces

Definition 2.1. A *topology* on a set X is a collection $\mathcal{T}$ of subsets of X having the following properties,

- (1) $\emptyset$ and X are in $\mathcal{T}$
- (2) The union of the elements of any subcollection of $\mathcal{T}$ is in $\mathcal{T}$
- (3) The intersection of the elements of any finite subcollection of $\mathcal{T}$ is in $\mathcal{T}$

A set X for which a topology $\mathcal{T}$ has been specified is called a topological space.

Note. A topological space is an ordered pair $(X, \mathcal{T})$

Remark. Note that properties (2) and (3) are closely related to the properties of open sets in analysis. In fact this is how we define open sets more generally.

Definition 2.2. If X is a topological space with topology $\mathcal{T}$ then a subset U of X is an **open set** of X if U belongs to the collection $\mathcal{T}$.

Remark. This is a definition for open sets in the context of a topology.

Note. Using this terminology we can say that a topological space is a set X with a collection of subsets of X called open sets such that $\emptyset$ and X are both open and such that arbitrary unions and finite intersections of open sets are open.

Example. If X is a set, the collection of all subsets of X is a topology on X called the *discrete topology*. The collection with just $\emptyset$ and X is called the *trivial topology* $\diamond$

Example. If X is a set and $\mathcal{T}_f$ is the collection of all subsets U of X such that $X - U$ is either finite or all of X , then $\mathcal{T}_f$ is a topology on X called the *finite complement topology*. First we have $X, \emptyset$ is in $\mathcal{T}_f$, since $X - X$ and $X - \emptyset$ are finite. Now given an indexed family of elements of $\mathcal{T}_f$ i.e. $\{U_\alpha\}$ that satisfy this property we need to show (2) and (3) hold. We see,

$$X - \bigcup U_\alpha = \bigcap (X - U_\alpha)$$

This is finite as each $X - U_\alpha$ is finite by assumption of U . Similarly given $U_1, \dots, U_n$ we see,

$$X - \bigcap_{i=1}^n U_i = \bigcup_{i=1}^n (X - U_i)$$

again the latter is finite as union of finite sets is finite. $\diamond$

Definition 2.3. Given $\mathcal{T}$ and $\mathcal{T}'$ are two topologies on X . If $\mathcal{T} \subset \mathcal{T}'$. Then $\mathcal{T}'$ is *finer* than $\mathcal{T}$. If $\mathcal{T}'$ properly contains $\mathcal{T}$ then it's *strictly finer*. We also use the terminology *coarser* and *strictly coarser* the other way around. $\mathcal{T}$ is *comparable* with $\mathcal{T}'$ if either $\mathcal{T} \subset \mathcal{T}'$ or $\mathcal{T}' \subset \mathcal{T}$

Note. This means that the "finest" topology is the discrete topology and the "coarsest" is the trivial.

2.2 Basis for a Topology

Definition 2.4. If X is a set, a *basis* for a topology on X is a collection $\mathcal{B}$ of subsets of X (called the basis elements) such that,

1. For each $x \in X$, there is at least one basis element B containing x .
2. If x belongs to $B_1 \cap B_2$, then there exists a basis element B_3 containing x such that $B_3 \subset B_1 \cap B_2$

Remark. If $\mathcal{B}$ satisfies the above two condition, then we define the *topology $\mathcal{T}$ generated by $\mathcal{B}$* as follows: A subset U of X is said to be open in X (in other words in $\mathcal{T}$) if for each $x \in U$, there is $B \in \mathcal{B}$ such that $x \in B$ and $B \subset U$. Each basis element is itself an element in $\mathcal{T}$.

Note this is basically saying that the topology generated by $\mathcal{B}$ is,

$$\mathcal{T} = \left\{ \bigcup \mathcal{C} \mid \mathcal{C} \subseteq \mathcal{B} \right\}$$

in other words we take all possible unions (including infinite unions) of the basis elements.

Example. If $\mathcal{B}$ is the collection of all circular regions in the plane (interiors of circles). Then $\mathcal{B}$ satisfies both conditions of a basis. In the topology generated, a subset U of the plane is open if every x in U lies in some circular region contained in U (note this implies that the border points cannot be included). $\diamond$

Example. If X is any set, the collection of all one-point subsets of X is a basis for the discrete topology on X . $\diamond$

We need to check that the collection $\mathcal{T}$ generated by $\mathcal{B}$ is a topology. First if $U = \emptyset$, then clearly it satisfies our condition and hence is in $\mathcal{T}$ if $U = X$ then clearly for each $x \in U$ there is a basis element B (by definition of a basis) containing X and trivially $B \subset X$ and hence $X \in \mathcal{T}$.

Now take two elements U_1, U_2 in $\mathcal{T}$. We need to show that $U_1 \cap U_2 \in \mathcal{T}$ or that for any $x \in U_1 \cap U_2$ there is some B such that $x \in B$ and $B \subset U_1 \cap U_2$. Clearly if $x \in U_1 \cap U_2$, $x \in U_1$ and $x \in U_2$ so we have B_1, B_2 such that $x \in B_1, B_2$ and $B_1 \subset U_1$ and $B_2 \subset U_2$ clearly $B_1 \cap B_2 \subset U_1 \cap U_2$ and (2) states that if $x \in B_1 \cap B_2$ we have B_3 s.t $x \in B_3 \subset B_1 \cap B_2 \subset U_1 \cap U_2$. Hence, $U_1 \cap U_2$ satisfies our condition for being in $\mathcal{T}$ and is open. Now, using induction we can show a finite intersection is also open.

Lastly, for arbitrary unions of an indexed family consider $\{U_\alpha\}_{\alpha \in J}$. Take $x \in \bigcup U$, this means for some α we have $x \in U_\alpha$ so exists a B such that $x \in B$ and $B \subset U_\alpha \subset \bigcup U$. Hence, exists B s.t $x \in B$ and $B \subset \bigcup U$ implies $\bigcup U \in \mathcal{T}$.

Lemma 2.5. If X is a set; and $\mathcal{B}$ a basis for a topology $\mathcal{T}$ on X . Then $\mathcal{T}$ equals the collections of all unions of elements of $\mathcal{B}$.

Proof. A collection of elements of $\mathcal{B}$ must be in $\mathcal{T}$ and union of elements in $\mathcal{T}$ are open, hence any union of basis elements must be in $\mathcal{T}$. Now for any $U \in \mathcal{T}$ for each $x \in U$ we can find B_i

s.t $x \in B_x, B_x \subset U$, hence we can write $U = \bigcup_{x \in U} B_x$ making it a union of basis elements. $\square$

Remark. Basically every open set U in X can be expressed as a union of it's basis elements, although not necessarily unique.

Note. This non-uniqueness in the representation of U w.r.t a basis is distinctive from the analogous basis vectors in linear algebra where the representation of a vector w.r.t a basis is unique (i.e the linear combination)

We can go from basis to a topology, we can also construct a basis given the topology itself.

Lemma 2.6. If X is a topological space. Suppose $\mathcal{C}$ is a collection of open sets of X such that for each open set U of X and each $x \in U$, there is an element $C \in \mathcal{C}$ such that $x \in C \subset U$. Then $\mathcal{C}$ is a basis for the topology of X .

Proof. We must show that $\mathcal{C}$ is a basis. We have two conditions to check. First clearly for any $x \in X$ we know that X is an open set i.e. $X \in \mathcal{T}$, from the lemma we know there is a $C \in \mathcal{C}$ such that $x \in C \subset X$ thence we satisfy the first condition of the basis.

Now let $x \in C_1 \cap C_2$ now since both C_1, C_2 are then $C_1 \cap C_2$ is also open and there is some $C_3 \in \mathcal{C}$ such that $x \in C_3 \subset C_1 \cap C_2$.

Now lastly, we need to show that the basis $\mathcal{C}$ generates the topology $\mathcal{T}$. Let $\mathcal{T}'$ be the topology generated by $\mathcal{C}$. Now let $U \in \mathcal{T}'$. By how we generate this means for any $x \in U$ there is some $C_x \in \mathcal{C}$ such that $x \in C_x \subset U$, but note that we have $C_x \in \mathcal{T}$ which means $U = \bigcup C_x \in \mathcal{T}$. Now on the other side if $U \subset \mathcal{T}$ then for each x we have $C_x \in \mathcal{C}$ such that $x \in C_x \subset U$ but trivially $C_x \in \mathcal{T}'$ and using the same logic we have $U \subset \mathcal{T}'$. Hence we get $\mathcal{T} = \mathcal{T}'$. $\square$

Lemma 2.7. Let $\mathcal{B}$ and $\mathcal{B}'$ be basis for the topologies $\mathcal{T}$ and $\mathcal{T}'$, on X . Then these are equivalent,

1. $\mathcal{T}'$ is finer than $\mathcal{T}$
2. For each $x \in X$ and each basis element $B \in \mathcal{B}$ containing x , there is a basis element $B' \in \mathcal{B}'$ such that $x \in B' \subset B$.

Remark. If a topological space is like a truckload of gravel. The pebbles can be thought of the basis elements; after the pebbles are smashed into smaller pieces, these pieces are the basis elements of the new topology which is obviously finer than the first.

2.3 The Order Topology

If X is a simply ordered set, then the standard topology generated using the order relation is called the order topology.

Definition 2.8. Let X be a set with a simple order relation with more than one element. And let $\mathcal{B}$ be the following collection of sets,

1. All open intervals (a, b) in X
2. All intervals of the form $[a_0, b)$ where a_0 is the smallest element of X .
3. All intervals of the form $(a, b_0]$ where b_0 is the largest element of X .

Then this collection is a basis for X and is called the **order topology**.

Remark. It is easy to see how this is a basis. For any $x \in X$ if x is the largest or smallest element then (2) and (3) ensures their existence. Else, (1) ensures it as we can choose (a_0, b_0) that will contain the element.

Similarly if x is in the intersection of any two sets, the intersection itself is of the form (1),(2) or (3) and hence is in the basis.

Example. Consider the order topology of the positive integers. In this case the topology generated is discrete (every singleton set is open) which means any subset of Z is also open. $\diamond$

Definition 2.9. If X is an ordered set, and a is an element of X , there are four subsets of X called the rays determined by a .

$$(a, +\infty) = \{x \mid x > a\}$$

$$(-\infty, a) = \{x \mid x < a\}$$

$$[a, +\infty) = \{x \mid x \geq a\}$$

$$(-\infty, a] = \{x \mid x \leq a\}$$

2.4 The Product Topology

If X, Y are topological spaces, then if we define a topology on $X \times Y$.

Definition 2.10. Let X and Y be topological spaces. The *product topology* on $X \times Y$ is the topology having as basis the collection $\mathcal{B}$ of all sets of the form $U \times V$ where U and V are open subsets of X and Y respectively.

Firstly $\mathcal{B}$ contains $X \times Y$ and hence all possible elements in $X \times Y$ is contained in some set in $\mathcal{B}$. For the second condition we have the intersection of any two basis element also forms a basis element as,

$$(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2)$$

and since $U_1 \cap U_2$ and $V_1 \cap V_2$ also are open.

Note. Note that $\mathcal{B}$ is not itself a topology as the union of two elements is not in the set. One can imagine two rectangles that overlap a bit, clearly the intersection forms a rectangle however the union doesn't.

Theorem 2.11. If $\mathcal{B}$ is a basis for X and $\mathcal{C}$ is a basis for Y then,

$$\mathcal{D} = \{B \times C \mid B \in \mathcal{B}, C \in \mathcal{C}\}$$

forms a basis for the topology of $X \times Y$.

Definition 2.12. Let $\pi_1 : X \times Y \rightarrow X$ be defined as,

$$\pi_1(x, y) = x$$

let $\pi_2 : X \times Y \rightarrow Y$ be defined as,

$$\pi_2(x, y) = y$$

then the maps π_1 and π_2 are called the **projections** of $X \times Y$ onto the first and second factors.

If U is an open subset of X , then the set $\pi_1^{-1}(U)$ is precisely the set $U \times Y$ which is open in $X \times Y$.

Theorem 2.13. The collection,

$$\mathcal{S} = \{\pi_1^{-1}(U) \mid U \text{ open in } X\} \cup \{\pi_2^{-1}(V) \mid V \text{ open in } Y\}$$

is a subbasis for the product topology on $X \times Y$

2.5 The Subspace Topology

Definition 2.14. Let X be a topological space with topology $\mathcal{T}$. If Y is a subset of X , the collection,

$$\mathcal{T}_Y = \{Y \cap U \mid U \in \mathcal{T}\}$$

is a topology on Y called the subspace topology. With this Y is called a subspace of X .

To show $\mathcal{T}_Y$ is a topology first we can choose $U = \emptyset$ which gives us $Y \cap U = \emptyset$ and choosing $U = X$ we get $U \cap Y = Y$ and hence we have $\emptyset$ and Y is in $\mathcal{T}_Y$.

Now for any two sets in $\mathcal{T}_Y$ say $Y \cap U_1$ and $Y \cap U_2$ we take the intersection as $Y \cap U_1 \cap U_2$ or $Y \cap (U_1 \cap U_2)$ and clearly $U_1 \cap U_2 \in \mathcal{T}$ as $\mathcal{T}$ is a topology and hence $Y \cap (U_1 \cap U_2)$ is in $\mathcal{T}_Y$. Taking arbitrary unions we get $Y \cap (U_1 \dots)$ and as $\mathcal{T}$ is a topology the arbitrary unions of U_i is open and hence $Y \cap (U_1 \dots)$ is in $\mathcal{T}_Y$.

Lemma 2.15. If $\mathcal{B}$ is a basis for the topology of X then the collection,

$$\mathcal{B}_Y = \{B \cap Y \mid B \in \mathcal{B}\}$$

is a basis for the subspace topology

Proof. First we show it's a basis. For any $y \in Y$ we have $y \in X$ and hence some B such that $y \in B$ which implies $y \in B \cap Y$ an element of $\mathcal{B}_Y$. Similarly if we have $y \in (B_1 \cap Y) \cap (B_2 \cap Y)$ then we have $y \in Y \cap (B_1 \cap B_2)$. This means $y \in B_1 \cap B_2$ and as $\mathcal{B}$ is a basis means exists B_3 such that $y \in B_3 \subset B_1 \cap B_2$. Hence we have $y \in Y \cap B_3 \subset Y \cap B_1 \cap B_2$.

Consider an open set U of X and consider an open set of Y under the subspace topology i.e is of the form $U \cap Y$. Note that we can choose an element B of $\mathcal{B}$ such that $y \in B \subset U$ (as $\mathcal{B}$ is a basis). So we also have $y \in B \cap Y \subset U \cap Y$. Which makes the above $\mathcal{B}_Y$ a basis for the subspace topology on Y . $\square$

Lemma 2.16. If Y is a subspace of X and if U is open in Y and Y is open in X then U is open in X .

Proof. If U is open in Y then exists V an open set in X such that $U = V \cap Y$. Now finite intersections of open sets are open. Hence U is open in X . $\square$